Conference Abstract

AI-Driven Emergency Care Coordination with Syndromic Surveillance (Urban ED Setting)

Author: Katharina, MD

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Background

Emergency departments (EDs) face fragmentation across handovers (EMS → triage → ED physician → specialty teams), manual orchestration of diagnostics, and dynamic resource constraints (beds, ICU, equipment). These frictions increase delays, cognitive load, and risk. We developed and field-tested a research pipeline that combines clinical gate logic, an existing MLP classifier, and a new syndromic surveillance layer trained on de-identified MIMIC-IV-ED data.

Methods

Data: MIMIC■IV■ED demo tables (edstays, triage, diagnosis, vitalsign; optional medrecon/pyxis), all de■identified.

Features: hourly arrivals; occupancy (interval expansion of stay windows); demographics (gender counts, age stats); triage/acuity distribution; chief■complaint token counts (top■K vocabulary); vitals aggregates (median/mean); harmonic day■of■week and hour encodings.

Models: (1) Production MLP gate engine for care prioritization (pre■existing).

(2) New unsupervised syndromic layer: SimpleImputer → StandardScaler → IsolationForest. The detector learns baseline hourly patterns and flags outlying surges (e.g., respiratory, GI) using keyword buckets and seasonal adjustment. Real■time scoring uses a one■hour sliding window assembled from the same feature builder to guarantee column alignment.

Governance: MIMIC-only, de■identified inputs; logging restricted to pseudonymous IDs and hourly buckets; explicit PHI/GDPR safeguards. We transparently used LLM assistance for code refactoring, error handling, and documentation under physician oversight.

Results

End■to■end runs completed with consistent feature extraction and anomaly scoring. The system generated plausible alerts (e.g., GI-syndrome warnings) while keeping the MLP gate engine intact. Smoke tests validated: bundle loading, gating, event logging, and real■time scoring. The architecture integrates into a clinician■facing UI without code changes to existing gates.

Conclusion

A modular, non■intrusive syndromic layer can augment ED coordination by surfacing early pattern shifts while respecting privacy and workflow boundaries. Future work: prospective evaluation, alert calibration to local patterns, fairness audits, and operational cost/benefit analysis.

Transparency statement

This abstract and associated codebase were produced by a physician (domain expert) with transparent use of LLMs for drafting, code suggestions, and debugging. All outputs were verified and executed on de■identified data only.